CDFD Runtime Paper 10: Multi-Domain Isomorphism and Adapter Evidence

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Abstract

Executive synthesis. Multi-domain isomorphism is the claim that many systems can be mapped into a common flow-constraint grammar. The current runtime exposes 196 registered domain adapters through the CLI. This is evidence of software breadth, not empirical proof that all domains obey CDFD. This paper explains the adapter contract and connects it to the finance-climate, universal-collapse, drug-synergy, and frontier-sweep experiments.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η, λ, μ, ν, ρ, σ, τ, and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 The Adapter Contract

Every domain adapter answers three questions:

1. What is the domain expression of flow Φ?
2. What is the domain expression of constraint C?
3. What interpretation is justified for $\Psi_s = (\Phi/C)SM_s$?

If these questions cannot be answered with measurable proxies, the adapter is not scientifically mature.

3 Runtime Breadth

The command

```
python cdfd.py domains
```

currently lists 196 registered domain keys (May 2026 inventory). Domains include medicine, physics, origins of life, ecology, economics, climate, infrastructure, materials science, software engineering, and many more.

This breadth is useful for hypothesis generation. It does not replace domain-specific validation. The `origins_of_life` adapter exposes Part II aromatic source-mix diagnostics through `cdfd.py demo origins_of_life --source-scenario <name>`.

4 Case Study: Finance and Climate

`experiments/notebooks/discovery_finance_climate.py` models a coupled toy system and tests whether financial Ψ volatility precedes increased climate constraint. The report presents a candidate directed edge. The key control is to remove coupling. If the edge persists after decoupling, the result is an artifact.

5 Case Study: Universal Collapse

`experiments/run_universal_discovery.py` models a central overloaded node in a network. It records hub Φ, hub C, hub Ψ, system mean Ψ_s, and collapsed-node count in `experiments/outputs/universal_collapse.json`. This is a useful generic cascade diagnostic for power grids, markets, organ systems, and transport networks, but each real domain still requires its own data mapping.

6 Case Study: Pharmacological Synergy

`experiments/run_drug_synergy_predictions.py` compares same-axis perturbations with cross-axis perturbations. In CDFD terms, one drug lowering Φ and another improving recovery or relaxation may behave differently from two drugs both lowering Φ. This is a toy pharmacology diagnostic, not a prescribing rule.

7 Frontier Sweep as Adapter Triage

`experiments/outputs/frontier_sweep.h5` records 2,500 parameter points and 2,487 non-trivial candidate records. The proper use is triage: identify which parameter regions deserve deeper testing, not declare thousands of laws.

8 Failure Conditions

Multi-domain isomorphism fails locally if:

- the adapter cannot define measurable Φ and C;
- predicted thresholds do not reproduce;
- the model performs no better than domain baselines;
- the same parameter story is forced onto incompatible systems.

9 Conclusion

The adapter library is an engine of disciplined analogy. It lets one runtime ask similar questions across many domains. The scientific standard remains local: each adapter must earn its interpretation with data, reproducible diagnostics, and falsification.

References

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics*. Public release archive, 2026, `https://doi.org/10.5281/zenodo.20264779`.